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Project: MD-Gooey
Period Covered: 9/29/2026 – 10/13/2026

Accomplishments since last report:

- Migrated frontend site over to regular path routing (createBrowserRouter) vs hash routing (createHashRouter); deployed test version.
    - Switched router
    - Added custom 404.html script to allow reloads
- Planned new local storage implementation to support simulation drafts *(high level implementation plan below):*
    1. *Move simulation drafts from session storage to local storage so they persist across sessions.*
    2. *Implement page for users to view drafts.*
- Researched Neon's OAuth library (@neondatabase/auth), which we will use to integrate Authentication the frontend.
- Completed implementation of mock Database & API wrapper with *Neon*, *Vercel*, and *node:http with Typescript*.
    - *Created database tables for current simulations and saved history (currently empty)*
    - *Added validation for required settings and supported force-field/water-model combinations*
    - *Added API input checks to reject invalid settings and prevent box edits from changing unrelated settings*
    - *Added 13 automated tests (all passing)*
    - *Documented local development and Vercel Preview setup*
    - *Imported repository into Vercel, added environment variables, deployed API; deployed health check passed.*
- Researched and started implementation for Multi-Site Deployment for GitHub Pages

Scheduled tasks to be done by next report:

- Deploy development version of web app on a sub route of main deployment, with github pages. *(10/17)*
- Fully functional database & api-layer, complete with all necessary provisions for authentication *(10/23)*
- First iteration of the Daimon script, connected to db with api layer. *(10/25)*
- Set up LMU (Microsoft) authentication on the development site, appropriately gating various frontend components & providing necessary visual cues to support authenticated vs unauthenticated use of the web app. Connect authentication to the Neon database (Neon's Managed Better API for authentication integrated with OAuth, specifically LMU/Microsoft). *(10/28)*

Noteworthy risks, concerns, or problems:

- Our Daimon is rejected by ITS due to some insecurity or other - to mitigate this we will run our Daimon by Brian Drawert and make sure everything checks out.
- Our Daimon damages important files on the workstation - this will also be mitigated by ITS, and by us running our Daimon by Brian.
- A bad actor accessing the workstation - this is mitigated by our entire database/daemon setup, and in addition Dr. Alavi is the only one who will be able to access the workstation directly.
- Currently no concrete design plans to authorize users to request simulation queuing or result viewing. Need to meet with the client (Dr. Alavi) to verify how many and what types of users should have access to the software.